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Cassette for dispensing pleated tubing

Abstract

A cassette to dispense a pleated tubing. The cassette includes a circular body having an annular channel housing and a pull tab and pull strip. At least one wall has an upper end with a lip disposed thereon. An annular cover is attached to the lip at the upper end defining a gap over the annular body. A tubing is disposed within the annular channel housing and withdrawn from the gap. At least one of a plurality of apertures is provided at a lower end of the annular channel housing. The apertures are adapted to be aligned with a projection extending upward from within a container.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. Design patent application Ser. No. 29/893,055, filed May 24, 2023; this application is also a continuation-in-part of U.S. patent application Ser. No. 17/889,367, filed Aug. 16, 2022; which is a continuation of U.S. patent application Ser. No. 17/066,349, filed Oct. 8, 2020, now U.S. Pat. No. 11,414,266; which claims priority to U.S. Provisional Patent Application Ser. No. 62/912,567, filed Oct. 8, 2019; and is a continuation-in-part of U.S. patent application Ser. No. 16/542,015, filed Aug. 15, 2019, now U.S. Pat. No. 11,130,628; which is a continuation of U.S. patent application Ser. No. 16/041,689, filed Jul. 20, 2018, now U.S. Pat. No. 10,435,235; which is a continuation of U.S. patent application Ser. No. 14/939,588, filed Nov. 12, 2015, now U.S. Pat. No. 10,053,282; which claims priority to U.S. Provisional Patent Application Ser. No. 62/078,915,

filed Nov. 12, 2014; and is a continuation-in-part of U.S. patent application Ser. No. 14/736,192, filed Jun. 10, 2015, now U.S. Pat. No. 10,486,925; which is a continuation of U.S. patent application Ser. No. 13/688,139, filed Nov. 28, 2012, now U.S. Pat. No. 9,085,404; which is a continuation of U.S. Design patent application Ser. No. 29/435,445, filed Oct. 24, 2012, now U.S. Pat. No. D695,541; the content of all of which are hereby incorporated by reference herein in their entirety into this disclosure.

TECHNICAL FIELD

(1) The subject disclosure relates to a cassette used for dispensing pleated tubing. More specifically, to a cassette capable of storing a pleated tubing and adapted for use within a disposal container to collect waste refuse.

BACKGROUND

(2) Various refillable cassettes have been provided for the disposal of waste material. Conventional dispensers typically require cumbersome techniques overcome by the disclosure below. Despite the ineffectiveness of these conventional attempts to provide a storage cassette, a need exists for a low cost, efficient storage container that can be conveniently assembled.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) Various exemplary embodiments of this disclosure will be described in detail, wherein like reference numerals refer to identical or similar components or steps, with reference to the following figures, wherein:

(2) FIG. 1 illustrates an upper perspective view of an exemplary cassette according to the subject disclosure.

(3) FIG. 1A shows an alternative upper perspective view of a cassette with a different engagement groove configuration.

(4) FIG. 2 depicts an exploded upper perspective view of an annular cover and annular body of the cassette according to the subject disclosure.

(5) FIG. 3 illustrates a lower perspective view of the cassette.

(6) FIG. 4 depicts an exploded lower perspective view of the annular cover and annular body of the cassette.

(7) FIG. 4A depicts a partial exploded lower perspective view of a section of the annular cover as shown in FIG. 4.

(8) FIG. 5 shows a side view of the cassette.

(9) FIG. 6 illustrates a bottom view of the cassette.

(10) FIG. 7 depicts a top view of the cassette.

(11) FIG. 8 illustrates a cross section side view of the annular cover and annular body of the cassette at through holes of the annular cover about line A-A in FIG. 7.

(12) FIG. 9 depicts a cross section side view of the annular cover and annular body of the cassette at an access hole and recessed plateau of the annular cover about line B-B in FIG. 7.

(13) FIG. 10 shows a cross section side view of the annular cover and annular body of the cassette at an upper horizontal wall about line C-C in FIG. 7.

(14) FIG. 11 illustrates a partial cross section side view of the annular cover and annular body of the cassette as shown in FIG. 8.

(15) FIG. 12 depicts a partial cross section side view of the annular cover and annular body of the cassette as shown in FIG. 9.

(16) FIG. 13 shows an exploded cross section side view of the annular cover and annular body of the cassette.

(17) FIG. 14 illustrates a cross section view of the cassette having a flexible tubing disposed

therein.

(18) FIG. 15 depicts a top perspective view of an alternate exemplary embodiment of the cassette.

(19) FIG. 15A illustrates a cross section view of the cassette fitted with the annular cover and having a flexible tubing disposed therein.

(20) FIG. 15B illustrates a top view of the cassette fitted with the annular cover.

(21) FIG. 16 depicts a second top perspective view of the cassette shown in FIG. 15.

(22) FIG. 17 shows a third top perspective view of the cassette from FIG. 15 without an annular cover or pleated flexible tubing inside.

(23) FIG. 18 illustrates a partial top perspective view of the cassette from FIG. 17 from an alternate viewing angle.

(24) FIG. 19 shows a bottom view of the cassette from FIG. 17.

(25) FIG. 20 illustrates a side perspective of the cassette from FIG. 17 that shows the bottom of the cassette.

(26) FIG. 21 shows a cross-section view of the cassette about line A-A in FIG. 19.

(27) FIG. 22 is an exploded view of the cassette shown in FIG. 21.

(28) FIG. 23 shows a cross-section view of the cassette about line B-B in FIG. 19.

(29) FIG. 24 is an exploded view of the cassette shown in FIG. 23.

(30) FIG. 25 is an exploded view of the cassette shown in FIG. 20.

(31) FIG. 26 is an exploded view of the cassette shown in FIG. 19.

(32) FIG. 27 is an exploded view of the cassette shown in FIG. 17.

(33) FIG. 28 is an exploded view of the cassette shown in FIG. 18.

(34) FIG. 29 is a top perspective view of the inside of a complementary waste container.

(35) FIG. 30 is a top view of the empty cassette of FIG. 17 installed in the complementary waste container.

(36) FIG. 31 is an exploded top view of the cassette of FIG. 17 and a mating adapter ring of the complementary waste container.

(37) FIG. 32 is an exploded side view of the cassette of FIG. 17 and the mating adapter ring of FIG. 31.

(38) FIG. 33 is a side view of the cassette of FIG. 17 mated with the adapter ring of FIGS. 31-32.

(39) FIG. 34 is a top view of the mated cassette and adapter ring of FIG. 33.

(40) FIG. 34A is a cross section of the cassette fitted with the annular cover and mated with the adapter ring.

(41) FIG. 34B is a top view of the cassette fitted with the annular cover and installed on the adapter ring.

(42) FIG. 35 is a top perspective view of a cassette with tubing and an annular cover installed in the complementary waste container.

(43) FIG. 36 depicts a top perspective view of another alternate exemplary embodiment of the cassette.

(44) FIG. 37 illustrates a bottom perspective view of an alternate exemplary embodiment of the cassette.

(45) FIG. 38 illustrates a top view of the cassette fitted with the annular cover.

(46) FIG. 39 illustrates a bottom view of the cassette fitted with the annular cover.

(47) FIG. 40 illustrates a side view of the cassette fitted with the annular cover.

(48) FIG. 41 illustrates another side view of the cassette fitted with the annular cover.

(49) FIG. 42 illustrates a cross-sectional view of the cassette fitted without the annular cover.

(50) FIG. 43 illustrates a cross-sectional view of the cassette fitted with the annular cover.

(51) FIG. 44 illustrates a close-up cross-sectional view of the cassette fitted without the annular cover.

(52) FIG. 45 illustrates a close-up cross-sectional view of the cassette fitted with the annular cover.

(53) FIG. 46 illustrates a close-up cross-sectional view of the cassette fitted without the annular

cover.

(54) FIG. 47 illustrates a close-up cross-sectional view of the cassette fitted with the annular cover and housing a pack of tubing.

(55) FIG. 48 illustrates use of the cassette with pull tab.

(56) FIG. 49 illustrates use of the cassette with pull tab.

DETAILED DESCRIPTION

(57) Particular embodiments of the present invention will now be described in greater detail with reference to the figures.

(58) FIGS. 1-4 illustrate upper and lower perspective and exploded perspective views of an exemplary cassette 10 according to this subject disclosure.

(59) FIGS. 1-4 show the cassette 10 is comprised of a lower annular body 20 having a generally U-shaped cross-section compartment and an annular cover 40 that extends over a portion of the U-shaped channel cross-section compartment. An opening 58 is provided between an outer concentric edge 49 of the annular cover 40 and an upper end 26 of an outer wall 24 of the lower annular body 20. The annular cover 40 is attached to the lower annular body 20 at a central cylindrical core 27, as will be discussed in greater detail below.

(60) FIGS. 2 and 13 illustrate an upper view of the cassette 10 having the annular cover 40. The annular cover 40 generally has an inner cylindrical wall 41 and upper horizontal wall 46. The annular cover 40 may have at least one recessed plateau 50 and at least one through hole 56 disposed in the upper horizontal wall 46. The annular cover 40 may also have at least one engagement groove 70.

(61) FIG. 2 shows that the annular cover 40 may be constructed with a series of vertical, angled and horizontal walls from the inner cylindrical wall 41 to the outer concentric edge 49 (as shown in greater detail in the cross-section views in FIGS. 11-12). The inner cylindrical wall 41 is connected to a first horizontal wall 42, which radially extends outward and terminates at a second cylindrical wall 43. The second cylindrical wall 43 is connected to a second horizontal wall 44, which radially extends outward and terminates at an upward angular wall 45. The upward angular wall 45 is connected to the upper horizontal wall 46, which terminates at a downward angular wall 47. The downward angular wall 47 extends downward for a predetermined distance and then curves upward at a curved portion 48 and is connected to the outer concentric edge 49 that extends upward to an edge 49a.

(62) The inner cylindrical wall 41, first horizontal wall 42, second cylindrical wall 43, second horizontal wall 44, upward angular wall 45, upper horizontal wall 46, downward angular wall 47, curved portion 48 and outer concentric edge 49 collectively may form the annular cover 40. It is to be understood, however, that the annular cover 40 may be constructed by any combination of the aforementioned segments to provide a cover of the U-shaped channel cross-section compartment.

(63) FIG. 3 depicts a lower perspective view of the cassette 10. The lower annular body 20 includes an inner wall 21 connected to an angular wall 22. The angular wall 22 is connected to a bottom wall 23, which is connected to an outer wall 24. The outer wall 24 terminates at an upper end 26 thereof. The inner wall 21, angular wall 22, bottom wall 23 and outer wall 24 collectively form the U-shaped channel cross-section of a housing into which a pack 82 of a pleated flexible tubing 80 is received, as shown in FIG. 14.

(64) The lower configuration of the U-shaped channel cross-section and/or the angular configuration taken by the angular wall 22 may take a variety of different suitable angles in order to allow air to escape from below during the packing of the flexible tubing 80 into the lower annular body 20 as a packed tubing 82 as shown in FIG. 14 and described in more detail later. For example, the angular wall may be directly connected between the outer wall 24 and the inner wall 21, without the need for a bottom wall 23.

(65) FIG. 3 illustrates the bottom wall 23 of the annular body 20 may also have a bottom lip 31, a plurality of concentric downward projections 32 and a plurality of apertures 34. The bottom lip 31

may be an extension of the outer wall **24** which extends past the bottom wall **23**. The bottom lip **31** and downward projections **32** may provide an alignment feature for stacking multiple cassettes **10**. A diameter of the bottom lip **31** may be sized such that when a first cassette **10** is stacked on the top of a second cassette **10**, either the inside or outside face of the bottom lip **31** will nest the upper end **26** of the annular body **20**. Furthermore, the concentric downward projections **32** may also provide a nesting function such that, the upper end **26** of the second cassette **10** is nested in between the bottom lip **31** and the concentric downward projections **32** or is nested inside the downward projections **32**.

(66) FIGS. **3-6** illustrate the plurality of apertures **34** that may be disposed in a radial configuration in the lower end of the annular body **20**. As shown, the apertures **34** may be elongated, radially extending inwardly lengthwise from a first end **34a** disposed in the bottom wall **23**, to a second end **34b** inwardly extending adjacent to the intersection of the angular wall **22** and the inner wall **21**. The apertures **34** may be cut into the angular wall **22** and the bottom wall **23** and disposed concentrically about in a radial pattern.

(67) The apertures **34** provide various advantages. First, during installation of the air-tight packing of the flexible packed tubing **82** into the U-shaped lower annular body **20**, the various apertures **34** serve as vent holes allowing air trapped below the packed tubing **82** to vent out of the lower annular body **20** through the apertures **34**. The venting provided by the various apertures **34** allow the packed tubing **82** to be compressed tightly as a pleated mass within the U-shaped lower annular body **20** without air interfering with the volume within the lower annular body **20** that could otherwise be filled by the packed tubing **82**. As a result, no air is trapped below the packed tubing **82** thereby allowing a tighter pack to be obtained so that more of the flexible tubing **80** in the compressed packed tubing **82** state can be stored within the lower annular body **20** during assembly of the tubing **80** into the cassette **10**. It is understood that the aperture **34** may be constructed into any one, or more, of the various walls **21**, **22**, **23**, **24**, or the like.

(68) Another significant advantage to the apertures **34** is the ability to control the rotation of the cassette **10**. The apertures **34** may function as key holes into which a mating key of a rotation mechanism can be used to control the rotation of the cassette **10** during operation of a unit (such as a waste receptacle) into which the cassette **10** may be placed and used. That is, a key may be aligned to mate with at least one of the apertures **34**. The key may engage any portion of the aperture **34** on any wall **21**, **22**, **23**, **24** surface and cause the cassette **10** to rotate, or prevent the cassette **10** from rotating by arresting the movement of the cassette **10**.

(69) Although the apertures **34** are shown as equidistant symmetric elongated rectangular slots extending across the angular wall **22** and the inner wall **21**, it is possible to vary the number of apertures **34**, their placement, the size and/or shape of the various apertures **34** to any number, size, symmetry or shape according to this subject disclosure. Likewise, is it also possible to extend the aperture **34** into the outer wall **24**, or alternatively provide the apertures **34** on any one, or more, of the inner wall **21**, the angular wall **22**, the bottom wall **23** or the outer wall **24**.

(70) FIGS. **4** and **4A** depict an exploded and enlarged lower perspective view of the annular cover **40** and annular body **20** of the cassette **10**. An inner surface of the inner cylindrical wall **41** may have a projection **52** which engages the annular body **20** to attach the annular cover **40** to the annular body **20**, as will be discussed in greater detail below.

(71) The annular cover **40** may also have a series of reinforcing bridges **36** along an underside thereof. An axis of each reinforcing bridge **36** may extend outward radially from a center of the annular cover **40**. The reinforcing bridges may span from the inner cylindrical wall **41** to the outer concentric edge **49**, or any portion thereof. The reinforcing bridges **36** structurally support the annular cover **40** to retain its shape as the flexible tubing **80** is pulled through the opening **58** over and through the center of the cassette **10**.

(72) FIGS. **7-13** depict various top views of the lower annular body **20** and the annular cover **40**. The annular cover **40** is disc shape and has a central through hole defined by an inner cylindrical

wall **41**. The inner cylindrical wall **41** extends upward to a first horizontal wall **42**. The first horizontal wall **42** extends from a first end inward to a second cylindrical wall **43**. The second cylindrical wall **43** extends upward to a second horizontal wall **44** that is elevated above the first horizontal wall **42**. At the lower end of the second cylindrical wall **43** a retaining flange or concentric shoulder **43a**, **43b** is formed that extends slightly below the lower surface of the first horizontal wall **42**. The second horizontal wall **44** extends radially outward to an upper angular wall **45**. The upper angular wall **45** extends radially upward and outward at an angle to an upper horizontal wall **46**. The upper horizontal wall **46** extends radially outward to a downward angular wall **47**. The downward angular wall **47** extends radially downward from a first end to a curved portion **48**. The curved portion **48** has a slight u-shape that returns the shape of the annular cover back upward into an outer concentric edge or wall **49**.

(73) The annular cover **40** is attached to the lower annular body **20** at an annular downturned lip **28** provided at a top edge **21a** of the inner concentric wall **21** defining the central cylindrical core **27**. The downward turned lip **28** constructed at the top edge **21a** of the central cylindrical core **27** is nestled within a cup shaped structure defined by the inner cylindrical wall **41**, the first horizontal wall **42** and the concentric shoulder **43a**, **43b** disposed at the bottom of the second cylindrical wall **43**. As shown in FIGS. **11-12**, the concentric shoulders **43a**, **43b** extend radially around and captivates the top edge **21a** of the inner concentric wall **21** to secure the annular cover **40** to the lower annular body **20**. It is to be understood that the inner cylindrical wall may take a variety of different configurations to provide additional support to the connection between the lower annular body **20** and the inner cylindrical wall **41** of the annular cover **40**. The thickness of the inner cylindrical wall **41** may be selectively thickened to provide a secure connection and to prevent the annular cover **40** from disengaging from the lower annular body **20**. Alternatively, bonding, an adhesive and/or any other suitable securing method may be applied between the annular cover **40** and the lower annular body **20** to ensure a secure connection.

(74) The shorter concentric shoulder **43a** may be extended intermittently and constructed with intermittent extending flanges **43b** or return walls, such as shown in exploded view in FIG. **4A**. The advantage of providing the intermittent extending flanges **43b** is to be able to more securely captivate and secure the top edge **21a** of the inner concentric wall **21** within the cup shaped structure defined by the inner cylindrical wall **41**, the first horizontal wall **42** and the concentric shoulder **43a**, **43b** disposed at the bottom of the second cylindrical wall **43**. The intermittent extending flanges **43b** may extend to, or at least beyond the width of the annular downturned lip **28** disposed at the top edge **21a** of the inner wall **21**. The intermittent extending flange **43b** may extend to at least the height of the protrusion **52b** disposed on the inside of the inner cylindrical wall **41**.

(75) The grooves **70** are adapted to receive an indexing mechanism to manipulate the orientation of the cassette **10**. For example, a user can manually use the grooves to engage with their fingers to manually twist the annular cover **40** into a desired position. The advantage to providing this feature is to manually rotate the cassette **10** via the top of the lower annular body **20**.

(76) By way of example shown in FIG. **1A**, the construction of the grooves **70**, **70a** may take a variety of different sizes, shapes and/or configurations according to this subject disclosure. The outer edge surface of the grooves **72** may be constructed of a material and/or shape having a low coefficient of friction so that the tubing pulled over those surfaces is not torn, frayed or fatigued by a sharp corner of edge. Likewise, the annular cover **40** may be constructed without the grooves **70**, **70a**.

(77) A plurality of apertures or through holes **56** may be provided on the top of the annular cover **40** for orientation purposes. Furthermore, the through holes **56** may be provided to allow aeration of a deodorant integrated with the material composition of the tubing material. The aeration through holes **56** can counteract any foul odor in the vicinity of the cassette **10**, particular when the cassette **10** used in a waste disposal container or unit.

(78) A plurality of adjacent elevated bosses and recessed plateaus may also be constructed around the upper surface of the annular cover **40**. The elevated bosses and recessed plateaus are constructed by a plurality of recessed plateaus being straddled by elevated ledges **51** that support various upper horizontal walls **46** forming a step-like configuration about the top end of the annular cover **40**. This construction has further advantages in permitting the cassette **10** to be rotated about the upper surface end of the annular cover **40**.

(79) As shown in FIGS. **11-12**, in place, the annular cover **40** and the lower annular body **20** are lockingly engaged to one another as described above. To prevent the annular cover **40** from being disconnected from within the expanded inner wall **21** of the lower annular body **20**, the annular cover **40** is lowered and positioned within an annular downturned lip **28** of the inner wall **21** of the annular body **20** so that a peripheral edge **28a** of the lip **28** slides past a ramp **52a** of the protrusion **52**. As the peripheral edge **28a** slides along the ramp **52a**, the inner wall **21** will deflect outward away from the center of the annular body **20** and against the concentric shoulders **43a**, **43b**. When the peripheral edge **28a** completely slides along the ramp **52a**, it will snap inward and rest above a locking shelf **52b** and against the concentric shoulders **43a**, **43b** as shown in FIGS. **11-12**.

(80) The peripheral edge **28a** of the annular downturned lip **28** is then locked against the locking shelf **52b** of the projection **52**. The projection **52** functions as a detent and the concentric shoulders **43a**, **43b** act as a captivating stop so that the annular cover **40** is mechanically arrested and cannot be undesirably lifted or raised off of the lower annular body **20** after the peripheral edge **28a** has been securely mounted against the locking shelf **52b** and the concentric shoulders **43a**, **43b**. In use, the flexible tubing **82** disposed within the cassette **10** is withdrawn with sufficient force upward from within the u-shaped lower annular body **20** and then over and downward over the annular cover **40** and into the cylindrical core **27** to cause the cassette to be jostled within the container it is situated. Unlike conventional designs before, the socket connection of this subject disclosure between the u-shaped lower annular body **20** and the annular cover **40** is sufficiently secure to prevent the shape of the cassette to be warped and the annular cover **40** disengaged from the lower annular body **20**.

(81) Access holes **54** are disposed in the first horizontal wall **42**. The access holes **54** are provided to allow visual alignment of the top edge **21** of the annular downturned lip **28** into the socket area constructed by the concentric shoulders **43a**, **43b**, the first horizontal wall **42**, the inner horizontal wall **41** and the locking flange **52**.

(82) FIG. **14** depicts the tubing **80** shown as a packed tubing **82** disposed in the U-shaped channel cross-section of the lower annular body **20**. The packed tubing **82** is adapted to be received and pulled upward from within the U-shaped channel, through the opening **58**, up and over the outer concentric edge **49**, across the annular cover **40**, and downward through the central cylindrical core **27** opening. The tubing **80** may be made from a variety of different materials. The tubing **80** may be made of various compositions and may also be scented.

(83) FIGS. **15-35** depict an alternate exemplary embodiment of the cassette **10** for dispensing pleated tubing. As shown in FIGS. **15-16**, the cassette **10** is fitted with an annular cover **40** that attaches to the lower annular body **20**. The lower annular body **20** has a lower closed channel cross-section comprising an inner wall **21** connected to a bottom wall **23** (shown in FIGS. **17** and **20**), which is connected to an outer wall **24**. The outer wall **24** terminates at an upper end **26** thereof. The inner wall **21**, bottom wall **23** and outer wall **24** collectively form the lower closed channel cross-section of a housing into which a pack **82** of a pleated flexible tubing **80** is received. FIGS. **15-18** show various tilted perspective views of the cassette **10** with and without the annular cover **40**. Two of a plurality of apertures **34** are shown disposed in the bottom of the lower annular body **20**. In tilted perspective view FIG. **15**, the apertures **34** are shown from an outer surface view of inner wall **21**. In tilted perspective views FIGS. **16-18**, the apertures **34** are illustrated from within the lower annular body **20**.

(84) FIG. **15A** is a cross section view of the cassette **10** fitted with the annular cover **40** and

containing the pack **82** of the pleated flexible tubing **80** in the lower annular body **20**. The outer wall **24**, bottom wall **23** and inner wall form the lower closed channel cross-section compartment that houses the pleated flexible tubing **80**. The outer wall **24** terminates at an upper limit at the upper end **26** of the lower annular body **20**. The cross-section is taken through two of the plurality of apertures **34**, while a third aperture **34** is also shown.

(85) FIG. **15B** is a top view of the cassette **10** fitted with the annular cover **40**. The annular cover **40** is secured to the lower annular body **20**.

(86) FIG. **17** is a top perspective view of the cassette **10** without the annular cover **40** or the pleated flexible tubing **80** that is housed in the internal compartment formed by the lower annular body **20**. The plurality of apertures **34** in the lower annular body **20** may be elongated, radially extending inwardly from a first end **34a** disposed in the bottom wall **23**, to a second end **34b** inwardly extending into the inner wall **21**. FIG. **18** illustrates a partial top perspective view of the cassette **10** from an alternate viewing angle that shows the plurality of apertures **34** disposed partially inside and outside of the inner wall **21**.

(87) FIG. **19** is a bottom view of the cassette **10**. In this embodiment, there are four apertures **34** set 90 degrees from each other. However, it is contemplated that a different number of apertures **34**, offset by different measures of degree may be adopted. For example, there could be three apertures **34** offset at 120 degrees, or the apertures **34** could be offset from one another in a non-symmetrical pattern.

(88) FIG. **20** depicts a side perspective of the cassette **10** that shows the bottom of the cassette **10**. The bottom wall of the annular body **20** may have a bottom lip **31**, a plurality of concentric downward projections **32** and a plurality of apertures **34**. The bottom lip **31** may be an extension of the outer wall **24** which extends past the bottom wall **23**. The bottom lip **31** may be used for stacking and storing more than one cassette **10**. The concentric downward projections **32** may also be used to orient and secure more than one cassette **10** when stacking or storing the cassettes **10**. The concentric downward projections **32** also add structural support to the bottom wall **23** of the cassette **10**. This added structural support is especially useful when loading the pleated flexible tubing **80** into the cassette **10**.

(89) As shown in FIG. **20**, each individual aperture **34e** has a gusset shape structure. That is, each individual aperture **34e** has a first aperture sidewall **135**, a second aperture sidewall **136** extending between the bottom wall **23** and the inner wall **21**. A rear aperture wall **138** is disposed at a first end **34a** of the aperture **34e** and connects the first aperture sidewall **135** and the second aperture sidewall **136** at the first end **34a**. The first aperture sidewall **135** and second aperture sidewall **136** extend from the first end **34a** of the aperture **34e** to a second end **34b** of the aperture **34e**. In the embodiment shown, the first end **34a** of the aperture **34e** is disposed in, and extending from the bottom wall **23** inward into the annular body **20**. The second end **34b** of the aperture **34** is disposed in the inner wall **21**. However, it is contemplated that the gusset structure of the aperture **34** may be arranged in a variety of different positions. That is, the first end **34a** and second end **34b** may be located entirely in either the outer wall **24**, the bottom wall **23**, the inner wall **21** or any combination thereof. An advantage of the aperture **34** having a gusset shape is that alignment and positioning of the cassette **10** within the waste container **150** is much easier to obtain as will be described in more detail later.

(90) FIG. **21** is a cross-section view of the cassette **10** about line A-A in FIG. **19**. The generally U-shaped compartment of the lower annular body **20** is comprised of the inner wall **21**, bottom wall **23** and outer wall **24**. Two of the plurality of apertures **34** extending to the second end **34b** are shown disposed in the inner wall **21**. FIG. **22** is an exploded view of the cassette shown in FIG. **21** that provides more detail of the bottom lip **31** and upper end **26** of the outer wall **24**.

(91) FIG. **23** is a cross-section view of the cassette about line B-B in FIG. **19**. This cross-section is taken through two of the plurality of apertures **34**. FIG. **24** is an exploded view of the cassette **10** shown in FIG. **23** that partially shows the architecture of a single aperture **34e**. The rear aperture

wall **138** is disposed on the first end **34a** of the aperture **34e**. The first aperture sidewall **135** and second aperture sidewall **136** extend from the first end **34a** of the aperture **34e** to the second end **34b** of the aperture **34e**. The rear aperture wall **138** may be disposed deeper towards the outer wall **24**, or shallower towards the inner wall **21**. The depth of the rear aperture wall **138** and the height of the first and second aperture sidewalls **135, 136** determine the angle of the opening of the aperture **34e**. The width of the bottom wall **23** affects the overall shape of the plurality of apertures **34**. It is contemplated that the first and second aperture sidewalls **135, 136** could be taller or shorter, and the rear aperture wall **138** deeper or shallower relative to inner wall **21**.

(92) FIGS. **25-28** show exploded views of the aperture **34e** from several different viewing angles. FIG. **25** shows an aperture **34e** from the perspective of one looking down at the bottom wall **23** of the cassette **10**. From this angle it is possible to see a closer view of the plurality concentric downward projections **32** disposed on the bottom wall **23**. The interior of the second aperture sidewall **135** and the rear aperture wall **138** are also shown.

(93) FIG. **26** is an exploded view of the cassette **10** from FIG. **19** looking down on the bottom of the cassette **10**. The contour of the aperture **34e** is shown. The rear aperture wall **138** is closest to the outer wall **24** at a first end **34a** of the aperture **34e** and connected by the first and second aperture sidewalls **135, 136** to a second end **34b** disposed in the inner wall **21**. In this embodiment the rear aperture wall **138** is wider than the second end **34b** of the aperture **34e** such that the lines of the first and second aperture sidewalls **135, 136** would bisect if continued into the central cylindrical core **27**. However, it is contemplated that the shape of the aperture **34e** could adopt different configurations such that the lines of the first and second apertures **135, 136** are parallel, or more acute or obtuse relative to the length of the rear aperture wall **138**.

(94) FIG. **27** is an exploded view of the cassette **10** shown in FIG. **17** looking down into the inside of the lower annular body **20** of an empty cassette **10**. This angle shows the exterior of the first aperture sidewall **135** and the interior of the second aperture sidewall **136**. The top edge of the rear aperture wall **138** is also shown.

(95) FIG. **28** is an exploded view of the cassette **10** from FIG. **18** that shows the exterior of the first aperture sidewall **135**, the interior of the rear aperture wall **138** and the interior of the second aperture sidewall **136**. In this embodiment, the shape of the first and second aperture sidewalls **135, 136** are quadrilateral. However, it is contemplated that the first and second aperture sidewalls **135, 136** may have rounded edges, parallel lines or more or less sides depending on the configuration of the complementary alignment projections **161** of the waste container **150**. Alternatively, the apertures may be substantially circular, semi-circular, elliptical, triangular, crescent-shaped, T-shaped, V-shaped, or the like, with one or more walls corresponding to the shape.

(96) FIG. **29** shows the complementary adapter ring **160** disposed in the interior of a waste container **150**. The adapter ring **160** is fitted with alignment projections **161** that mate with the plurality of apertures **34** of the cassette **10**. To accommodate the design of this particular waste container **150**, which has four alignment projections **161** offset at 90 degrees around a central axis **150A** of the waste container **150**, the complementary cassette **10** has four apertures **34**. The alignment projections **161**, and the corresponding apertures **34** in the cassette **10**, control, or prevent, the rotation of the cassette **10**.

(97) The internal architecture of the plurality of apertures **34** in a gusset shape is optimized for the structural integrity of the cassette **10**. The first aperture sidewall **135**, second aperture sidewall **136** and rear aperture sidewall **138** help locate the alignment projections **161** when installing the cassette. The first and second aperture sidewalls **135, 136** provide structural support against the alignment projections **161** during rotation or when preventing rotation. The first aperture sidewall **135**, the second aperture sidewall **136** and the rear aperture sidewall **138** also elevate the pleated flexible tubing **80** accumulated within the lower annular body **20** away from the outer surface of the lower annular body **20**. This elevation in the walls **135, 136, 138** of the aperture **34** is beneficial in substantially reducing the interaction or interference with the alignment projections **161** that are

aligned with and inserted into the apertures **34** such as shown in FIG. **33**. Thus, the alignment projections **161** can easily be aligned and registered within the apertures **34**. This elevated construction also helps facilitate the loading of the pleated flexible tubing **80** into the cassette **10** and the removal of the pleated flexible tubing **80** during use.

(98) FIG. **30** shows the empty cassette **10** from FIG. **17** installed in the waste container **150**. The alignment projections **161** mate with the plurality of apertures **34** located in the bottom of the cassette **10**. Once installed, the motion of the cassette **10** is determined by the rotation, or lack thereof, of the adapter ring **160**. The cassette **10** is installed flush against the adapter ring **160** with the alignment projections **161** extending through the plurality of apertures **34** of the cassette **10**. As shown in FIGS. **15-35**, the plurality of apertures **34** provide an open through-hole from the outside of the cassette that opens into the interior space of the annular body **20**.

(99) However, it is contemplated that the plurality of apertures **34** may not open into the interior of the lower annular body **20** and each individual aperture **34e** may also have an upper aperture wall **139** (not shown). The upper aperture wall **139** in each of the plurality of apertures **34** would surround the top of the alignment projections **161** and prevent the alignment projections **161** from protruding into the internal cavity of the lower annular body **20** as they do with the open through-hole design shown in FIGS. **15-35**.

(100) FIG. **31** is an exploded view of the empty lower annular body **20** from FIG. **17** and the mating adapter ring **160** of a complementary waste container **150**. The alignment projections **161** are lined up with the plurality of apertures **34** disposed in the bottom of the lower annular body **20**. When the cassette **10** is lowered onto the adapter ring **160** the plurality of apertures **34** may function as key holes into which a mating key or the alignment projections **161** can be used to control the positioning and/or rotation of the cassette **10** during operation of a unit (such as a waste receptacle) into which the cassette **10** may be placed and used. The alignment projection **161** may be adapted to engage any portion of the aperture **34** that can be provided on any wall **21**, **23**, **24** surface and cause the cassette **10** to rotate, or prevent the cassette **10** from rotating by arresting the movement of the cassette **10**.

(101) FIG. **32** is a side cross-section view of the cassette **10** about line B-B from FIG. **19** with the adapter ring **160** in position. FIG. **33** is an exploded view of the same. FIGS. **32-33** show the alignment projections **161** mating with the plurality of apertures **34** of the cassette **34**. In the embodiment shown, the alignment projections **161** of the adapter ring **160** have a hooked shape that engages with the rear aperture wall **138** of the plurality of apertures **34**. However, it is contemplated that the alignment projections **161** may adapt a variety of shapes and configurations that mate with the plurality of apertures **34** such as: triangular, rectangular, circular, semicircular, vertical, and/or various three-dimensional solids formed by extension through space of these and other suitable geometric shapes.

(102) FIG. **34** shows a top view of the cassette **10** installed on the adapter ring **160**. In this embodiment, the alignment projections **161** extend into the interior of the lower annular body **20**. However, the alignment projections **161** could fit within, or flush with the sidewalls **135**, **136** of the plurality of apertures **34**.

(103) FIG. **34A** is a cross-section of the cassette **10** fitted with the annular cover **40** and mated with the adapter ring **160**. The interaction between the plurality of apertures **34** and the alignment projections **161** is shown. The alignment projections **161** extend through the plurality of apertures **34** into the internal compartment of the lower annular body **20**.

(104) FIG. **34B** is a top view of the cassette **10** fitted with the annular cover **40** and mated with the adapter ring **160**. The alignment projections **161** are shown through the empty lower annular body **20** and the apertures disposed in the annular cover **40**.

(105) FIG. **35** shows a cassette **10** filled with pleated flexible tubing **80** and fitted with the annular cover **40** and installed in a waste container **150**.

(106) FIGS. **36-49** depict another alternate exemplary embodiment of a cassette **10** for dispensing

pleated tubing. As shown in FIG. 36, cassette 10 is fitted with an annular cover 40 that attaches to the lower annular body 20. The lower annular body 20 has a lower closed channel cross-section compartment comprising an inner wall 21 connected to a bottom wall 23 (similar to that shown in FIGS. 17 and 20), which is connected to an outer wall 24. The outer wall 24 terminates at an upper end 26 thereof. The inner wall 21, bottom wall 23 and outer wall 24 collectively form the lower closed channel cross-section of a housing into which a pack 82 of a pleated flexible tubing 80 is received (shown in FIGS. 46-47, and similar to that shown in FIG. 15A). Two of a plurality of apertures 34 are shown disposed in the bottom of the lower annular body 20. In the tilted perspective view FIG. 36, the apertures 34 are shown from an outer surface view of inner wall 21. (107) The annular cover 40 may also have at least one pull tab 90 having an outer edge 92 and positioned directly above a tab aperture 93. The pull tabs 90 may include direction arrows to guide the user to pull up and in a specific direction to remove from the lower annular body 20. The pull tabs 90 and pull strips 91 cover the gap through which the flexible tubing 80 is drawn out when the pull tabs 90 and the pull strips 91 are removed. The use of the pull tab 90 will be described in more detail in FIGS. 48-49.

(108) By way of example shown in FIG. 36, the construction of the tabs 90 may take on a variety of different sizes, shapes and/or configurations according to this subject disclosure. In the example shown in FIGS. 36 and 38, the tab 90 is trapezoidal with a longer side on an outer edge and a shorter side on an inner edge. But the tab 90 may also be rectangular, semi-circular, triangular or other shape as appreciated by one having ordinary skill in the art. The tabs 90 may have multiple layer construction, and be flexible enough to be readily removed by a user, as shown and described in FIGS. 48-49.

(109) FIG. 37 depicts a side perspective view of the bottom of cassette 10. The bottom wall 23 of the annular body 20 may have a bottom lip 31, a plurality of concentric downward projections 32 and a plurality of apertures 34. The bottom lip 31 may be an extension of the outer wall 24 which extends past the bottom wall 23. The bottom lip 31 may be used for stacking and storing more than one cassette 10. The concentric downward projections 32 may also be used to orient and secure more than one cassette 10 when stacking or storing the cassettes 10. The concentric downward projections 32 also add structural support to the bottom wall 23 of the cassette 10. This added structural support is especially useful when loading the pleated flexible tubing 80 into the cassette 10.

(110) As shown in FIG. 37, each individual aperture 34 has a gusset shape which is created by cutting out a rectangular portion of the inner wall 21 and a rectangular portion of the bottom wall 23. A first end 34a of the aperture 34 extends into the bottom wall 23 and a second end 34b of the aperture 34 extends into the inner wall 21. However, it is contemplated that the structure of the aperture 34 may be arranged in a variety of different positions. That is, the first end 34a and second end 34b may be located entirely in either the outer wall 24, the bottom wall 23, the inner wall 21, or any combination thereof. An advantage of the aperture 34 having a gusset shape is that alignment and positioning of the cassette 10 within the waste container 150, as well as maximizing the amount of tubing that may be stored in the lower annular body 20, is much easier to obtain as described elsewhere in this disclosure. A difference between the aperture 34 in this embodiment versus the prior embodiments is that there are no internal walls or structure to the present aperture 34, unlike those described in the prior embodiments (such as FIG. 25). In essence, the present aperture 34 shown in this embodiment is a "cut out" of the surfaces of the bottom wall 23 and inner wall 21.

(111) FIG. 38 is a top view of cassette 10 fitted with the annular cover 40. The annular cover 40 is secured to the lower annular body 20. The opposing tabs 90 and tab apertures 93 are positioned 180 degrees from each other. Rip gaps 25 at opposite ends (also shown in FIG. 36, and during use in FIGS. 48-49) provide an easy tear bond which allows the tabs 90 and attached pull strips 91 to be pulled concentrically off the top of the annular cover 40 before use, as will be described in more

detail with respect to FIGS. **48-49**.

(112) FIG. **39** is a bottom view of cassette **10**. In this embodiment, there are four apertures **34** set 90 degrees from each other. However, it is contemplated that a different number of apertures **34**, offset by different measures of degree may be adopted. For example, there could be three apertures **34** offset at 120 degrees, or the apertures **34** could be offset from one another in a non-symmetrical pattern. Also visible in this figure are the inner wall **21**, bottom wall **23**, and outer wall **24**.

(113) FIGS. **40** and **41** show top/bottom and side views, respectively, of cassette **10**. Shown in the figures are the rip gap **25** with tear bridge **29** depicting an annular lip **20A** therein, elevated portions of the tabs **90**, pull strips **91**, lower annular body **20**, and annular cover **40**.

(114) FIGS. **42-43** show cross-sectional views of just a cassette **10** and cassette **10** with a cover **40**, respectively, along plane K-K of FIG. **38**. Although the cross-sectional area is not exact, it is illustrative of the interior of cassette **10**. As compared to the cross-sectional views of other embodiments (see FIG. **13**, **21**, **23**, etc.), the present embodiment uses less material because apertures **34** do not have internal walls (like those shown in FIG. **25**) and a tear away opening including the pull tab **90** and the pull strips **91** are provided. Apertures **34** are simple cut outs of the inner wall **21** and bottom wall **23**, which allow direct access between an interior portion of the cassette **10** and the channel formed between the inner wall **21**, bottom wall **23**, and outer wall **24**. Thus, the present embodiment is easier and less costly to manufacture, and less prone to failure. Likewise, the simple cut outs allow more tubing to be packed into the lower annular body **20**.

(115) FIGS. **44-45** show close-up cross-sectional views of an end of the cassette **10** without a cover **40**, and with a cover **40**, respectively. FIGS. **46-47** show close-up cross-sectional views of an end of the cassette **10** without a cover **40**, and with a cover **40**, respectively, and housing a pack **82** of pleated flexible tubing **80**. The pleated tubing **80** is secured within cassette **10** and topped off with cover **40** so that it is ready for use. As shown in FIG. **45**, when the annular cover **40** is secured onto the lower annular body **20**, the annular cover lip **40A** protruding inward is locked onto the annular lip **20A** that protrudes outward against the annular cover lip **40A**. When the pull tabs **90** and the pull strips **91** are torn away from the annular cover **40**, the annular cover lip **40A** is torn away and disengaged from the annular lip **20A** to allow its removal.

(116) FIGS. **48-49** show the use of the pull tab **90** and the pull strips **90A** being torn away from the annular cover **40**. For sake of presenting a clearer description of use, in these figures, pull tabs **90** are further labeled as **90A** and **90B**, pull strips **91** are further labeled as **91A** and **91B**, and rip gaps **25** are further labeled as rip gap **25A** and rip gap **25B**. As shown in FIGS. **36** and **38**, pull tabs **90** may have a directional marking to indicate to the user which way to pull the tab **90** up and around the top portion of cassette **10**. As shown in FIGS. **36** and **48**, a user would insert a finger within aperture **93**, which is positioned directly underneath the tab **90**, and pull upwards, or radially outward, on pull tab **90A** to break the adjacent rip gap **25A**. The force of pulling upwards and/or radially outward, on pull tab **90A** results in the tear of bridge **29** (see FIG. **40**) in rip gap **25A** that connects the side of the adjacent pull strip **91** (shown as **91A** and **91B** in FIGS. **48-49**). As the user pulls up and clockwise on pull tab **90A**, as shown in FIG. **48**, the pull strip **91A** separates from the annular top **40** until pull strip **91A** reaches the second rip gap **25B**. The user continues the upward and clockwise pull up on the pull tab **90A**, as shown in FIG. **49**, so that the second pull tab **90B** is also pulled up with the pull strip **91A** attached to the first pull tab **90A** in the user's finger. The user continues to pull up on pull tab **90A** until both pull tabs **90A/90B** and both pull strips **91A/91B** are separated from the annular top **40**. The user then has access to the flexible tubing **80** to pull out and engage into use, as described above. The second rip gap **25B** is sufficiently strong enough not to tear as the pull tab **90A** is being pulled upward through the second pull tab **90B**. However, in case the first pull tab **90A** and first pull strip **91A** get detached from the second pull tab **90B**, then the user can pull up on pull tab **90B** and continue the upward and clockwise pull motion to separate pull tab **90B** and pull strip **91B** from the annular top **40**. Thus, there is a pull tab and pull strip system to open up the gap for the flexible tubing **80** to be accessed and used.

(117) The illustrations and examples provided herein are for explanatory purposes and are not intended to limit the scope of the appended claims. It will be recognized by those skilled in the art that changes or modifications may be made to the above described embodiment without departing from the broad inventive concepts of the invention. It is understood therefore that the invention is not limited to the particular embodiment which is described, but is intended to cover all modifications and changes within the scope and spirit of the invention.

Claims

1. A cassette to dispense a pleated tubing, comprising: a body having a channel housing formed by an outer wall, an inner wall, and a bottom wall; a cover attached to the body that defines a gap over the body; a pull tab attached to a pull strip and positioned atop the gap; a tubing disposed within the channel housing and withdrawn from the gap; and a plurality of apertures, of which at least one aperture having a gusset shape no internal walls projecting into the channel housing and extending from a first end at the bottom wall to a second end at the inner wall of the body adapted to align with and receive a projection extending upward.
2. The cassette to dispense a pleated tubing recited in claim 1, wherein the at least one aperture provides an opening through the lower end of the body.
3. The cassette to dispense a pleated tubing recited in claim 1, further comprising a plurality of pulls tabs and pull strips positioned in series and separated by a plurality of rip gaps and tear bridges.
4. The cassette to dispense a pleated tubing recited in claim 1, wherein the tubing is elevated to allow a mating projection to be inserted into the at least one aperture without substantial interference from the tubing.
5. The cassette to dispense a pleated tubing recited in claim 1, wherein the at least one aperture is disposed concentrically about the body and extends from at least one of the outer wall, the bottom wall or the inner wall.
6. The cassette to dispense a pleated tubing recited in claim 1, wherein the at least one aperture is adapted to mate with a keying mechanism disposed within a container to control the angular rotation of the cassette.
7. The cassette to dispense a pleated tubing recited in claim 1, wherein the at least one aperture is adapted to align with and receive the projection extending upward from a container.
8. The cassette to dispense a pleated tubing recited in claim 1, wherein the at least one aperture is adapted to align and receive a projection extending upward from an adapter ring disposed within a container.
9. The cassette to dispense a pleated tubing recited in claim 1, wherein a lip extends downwards from the outer wall past a point of intersection with the bottom wall, and wherein at least one concentric downward projection is disposed on the bottom wall.
10. A cassette to dispense a pleated tubing, comprising: a body having a channel housing formed by an outer wall, an inner wall, and a bottom wall; a cover defining a gap over the body; a tubing disposed within the channel housing and withdrawn from the gap; a plurality of pull tabs attached to a plurality of pull strips and positioned in series atop the gap; and a plurality of apertures, of which at least one aperture with a gusset shape is having no internal walls projecting into the channel housing, and disposed in a lower end of the body, and extending from a first end at the bottom wall to a second end at the inner wall of the body; wherein the at least one aperture provides an opening through a lower end of the body adapted to receive an alignment projection extending upward.
11. The cassette to dispense a pleated tubing recited in claim 10, wherein the at least one aperture is adapted to receive more than one alignment projections to control the rotation of the cassette.
12. The cassette to dispense a pleated tubing recited in claim 10, wherein the alignment projection

extends upward from a waste container.

13. A cassette to dispense a pleated tubing, comprising: a body having a channel housing; a cover attached to the body that defines a gap over the body; a tubing disposed within the channel housing and withdrawn from the gap; and a plurality of pull tabs attached to a plurality of pull strips and positioned in series atop the gap and separated by a plurality of rip gaps and tear bridges; and a plurality of apertures, of which at least one aperture having a gusset shape with no internal walls projecting into the channel housing, and disposed in a lower end of the body and adapted to receive a mating projection extending upward.

14. The cassette to dispense a pleated tubing recited in claim 13, wherein the at least one aperture provides an opening through the lower end of the body.

15. The cassette to dispense a pleated tubing recited in claim 13, wherein the at least one aperture extends from a first end at a bottom wall to a second end at an inner wall of the body.

16. The cassette to dispense a pleated tubing recited in claim 13, wherein the pleated tubing is elevated to allow the mating projection to be inserted into the at least one aperture without substantial interference from the pleated tubing.

17. The cassette to dispense a pleated tubing recited in claim 13, wherein the at least one aperture is disposed concentrically about the body and extends from at least one of an outer wall, a bottom wall or an inner wall.

18. The cassette to dispense a pleated tubing recited in claim 13, wherein the mating projection is a keying mechanism disposed within a container to control the angular rotation of the cassette.

19. The cassette to dispense a pleated tubing recited in claim 13, wherein the at least one aperture is adapted to align with and receive the mating projection extending upward from a container.

20. The cassette to dispense a pleated tubing recited in claim 13, wherein the at least one aperture is adapted to align and receive the mating projection extending upward from an adapter ring disposed within a container.
